Supplementary Material

This document is the supplementary material of the thesis. Its sections are Supplements A to C. Section, table, figure and equation numbers with a letter refer to this document; numbers without a letter refer to the thesis.

Supplement A Parameter Sweeps and the Response Rule

A.1 Sweep of the LLM Settings

To choose the LLM configuration before the main experiments, we ran a factorial sweep over five factors[1]: Prince agent type (LLM, random), kind of opponent (LLM, random), temperature (0.5, 0.7), top-p (0.8, 0.9) and reasoning history (enabled, disabled). This produced 32 configurations with 20 games each (640 games in total).[2]

The experiment used a simplified 5-player setting with no center card and only Princess-faction and Queen-faction roles.

In Figure A.1, every LLM setting has an effect below 6 points: 5.5 points for top-p and less than 3 points each for temperature and reasoning history. This suggests that the win rate is relatively robust to these settings within the tested ranges.[3]

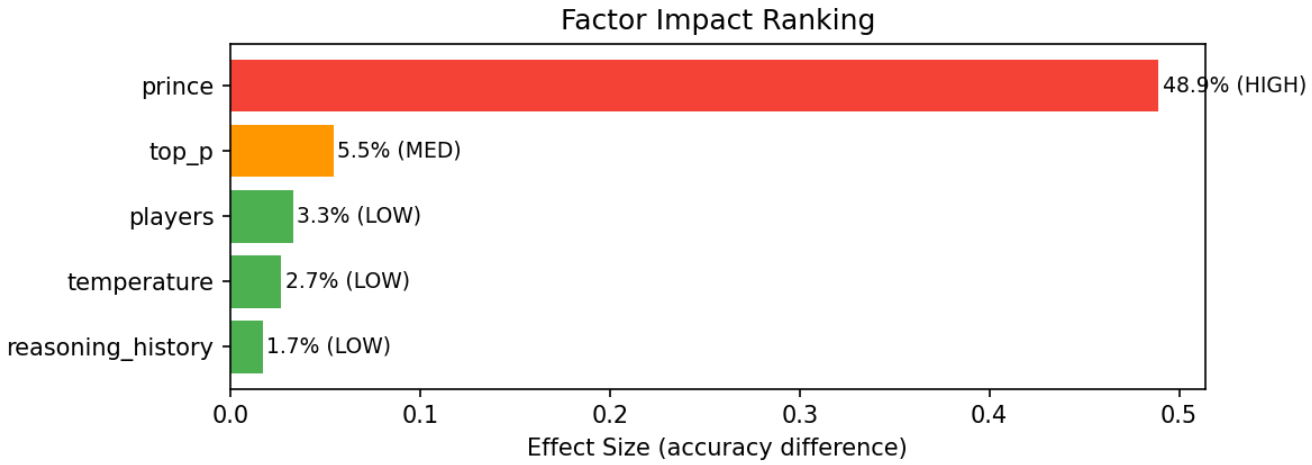


Figure A.1: Main effects of the five factors of the configuration sweep (5 players, no center card).

Among the four best runs of the sweep (10 games each, with the two repeat penalties ranked as separate runs), one played against random opponents. Random opponents are weaker and of a different kind, so that configuration is not comparable with the others. We validated the other three, all against LLM opponents, at full scale (7 players with center card, 100 games each). The top configuration (LLM vs LLM, reasoning history disabled, temperature 0.5, top-p 0.8) won 65% and averaged 1253 seconds per game. The second (LLM vs LLM, reasoning history enabled, temperature 0.5, top-p 0.9) won 63% and averaged 466 seconds per game. On the same 100 role assignments the two cannot be separated: 19 games are won only by the first and 17 only by the second (exact McNemar $p = 0.87$). The second is more than twice as fast for a win rate only 2 points lower. We therefore adopt it as the LLM Prince baseline for the rest of our experiments. Both are far above the random baseline, 14% in the same game setting with a different seed.

[1] This sweep was published during the project as [1].

[2] Since the Ollama API ignored the repeat penalty, the results are pooled over it.

[3] The effect of a factor is the difference between the mean win rates at its two levels. Figure A.1 averages the win rates of the 64 runs of 10 games, one for each configuration and repeat penalty, as the workshop paper [1] does. This gives 75.8% for the LLM Prince and 48.9 points for the Prince type. Pooling the games instead gives 75.9% for the LLM Prince, 26.9% for the random Prince and 49.0 points.

A.2 Sweeps of the Response Rule

We ran two further sweeps to choose the response rule of the level-1 and level-2 opponent models. The main sweeps use the Sym(lv.2) Prince on 5-player games, 50 games per cell, against level-0 or level-2 opponents, with and without a center card (Tables A.1 and A.2). Sweeps of the softmax sharpness β with 7 players (20 and 100 games per cell) are not reported.

In the sweep of the epsilon-greedy error rate ε from 0.01 to 0.5 (Table A.1), the four conditions do not agree. Against level-2 opponents without a center card, almost every value wins 100%. Against level-0 opponents without a center card, the win rate rises with ε, which may be because a large ε makes the level-1 model close to the level-0 model. We chose $\varepsilon = 0.2$, which has the best worst case, 78% against 74% for $\varepsilon = 0.5$. Its mean, 85.0%, is second only to the 87.0% of $\varepsilon = 0.5$.

Table A.1. Win rate of the Sym(lv.2) Prince with each epsilon-greedy error rate ε, with the mean and the worst case over the four conditions (5 players, 50 games per cell, base seed 65685).

ε	level-0, center card	level-2, center card	level-0, no center card	level-2, no center card	Mean	Worst case
0.01	64%	78%	66%	94%	75.5%	64%
0.02	60%	78%	64%	100%	75.5%	60%
0.05	74%	64%	70%	100%	77.0%	64%
0.10	70%	72%	70%	100%	78.0%	70%
0.20	80%	78%	82%	100%	85.0%	78%
0.50	90%	74%	84%	100%	87.0%	74%

The second sweep varies the softmax sharpness β from 0 to 128 (Table A.2). No sharpness is best in all conditions: a Prince with a sharper softmax wins more against level-2 opponents and less against level-0 opponents. This is consistent with the missing odds bound of softmax (Supplement A.3), under which one unexpected answer can make the weight of the true world close to zero. Of the sharpness values, $\beta = 16$ has both the highest mean, 85.0%, and the best worst case, 74%.

Table A.2. Win rate of the Sym(lv.2) Prince with a softmax rule for each sharpness β, with the mean and the worst case over the four conditions (5 players, 50 games per cell, base seed 65685).

β	level-0, center card	level-2, center card	level-0, no center card	level-2, no center card	Mean	Worst case
0	98%	62%	88%	46%	73.5%	46%
1	100%	54%	94%	52%	75.0%	52%
2	98%	50%	86%	50%	71.0%	50%
4	98%	48%	84%	48%	69.5%	48%
8	92%	68%	72%	100%	83.0%	68%
16	88%	78%	74%	100%	85.0%	74%
32	66%	86%	58%	100%	77.5%	58%
64	78%	76%	48%	100%	75.5%	48%
128	66%	60%	48%	100%	68.5%	48%

The runs with $\beta = 0$ against level-0 opponents are the only matched runs of the sweeps (the opponent model describes the opponents exactly), because the level-0 opponents follow the uniform model.

With the selected parameter of each rule fixed, the two rules have the same mean over the four conditions, 85.0% (Tables A.1 and A.2). Epsilon-greedy has the better worst case, 78% against 74%. We use epsilon-greedy based on these sweeps and on its odds bound, which Supplement A.3 derives.

A.3 Odds Bound of the Epsilon-Greedy Rule

The level-1 and level-2 models first used softmax, which is the default of the code (Supplement B). The odds bound limits how far one answer can move the belief. Equation A.1 states the epsilon-greedy rule, with R^* the set of $\ell \geq 1$ valid answers of highest value and K the number of valid answers of the opponent in w.

Equation A.1

$$P(r \mid w, q, c) = (1 - \varepsilon)\,\frac{\mathbb{1}[r \in R^*]}{\ell} + \frac{\varepsilon}{K}$$

Consider one answer r and two worlds w and w' in which r is valid for the answering opponent. The belief update (Equation 5.1) multiplies the odds between the two worlds by the likelihood ratio $P(r \mid w, q, c)/P(r \mid w', q, c)$, so a bound on this ratio bounds how far one answer can move the belief.

Under Equation A.1 every valid answer has probability at least ε/K, and the largest probability is $(1-\varepsilon)+\varepsilon/K$, reached when a single answer has the highest value. With ℓ tied answers the largest probability is $(1 - \varepsilon)/\ell + \varepsilon/K$, which is smaller. These two extremes are the same in every world with K valid answers, whatever the values are. The ratio is therefore at most the bound of Equation A.2, which is 29 for $K = 7$ and $\varepsilon = 0.2$.

Equation A.2

$$\frac{P(r \mid w, q, c)}{P(r \mid w', q, c)} \leq \frac{(1 - \varepsilon) + \varepsilon/K}{\varepsilon/K} = 1 + \frac{K(1 - \varepsilon)}{\varepsilon}$$

If the opponent has K valid answers in w and K' in w', the same argument gives $(1 - \varepsilon)K'/\varepsilon + K'/K$. A Queen-faction player has one valid answer fewer than an Other-faction player, so this is at most $28+7/6 \approx 29.2$ in the 7-player game. A Princess-faction player answers deterministically, so when w is such a world, $K = 1$ and the ratio can reach $K'/\varepsilon = 35$ with $K' = 7$. This is the bound that holds for every pair of worlds, since an answer has probability at most 1 in w and at least $\varepsilon/K' \geq \varepsilon/7$ in w'.

Under softmax, $P(r \mid w, q, c) \propto \exp(\beta\, v(r))$, the ratio between the most and the least likely answer in one world is $\exp(\beta\,(v_{\max} - v_{\min}))$. It grows without limit as β grows, for any fixed gap between the values, so no bound holds for every sharpness.

The sweeps cannot isolate this bound. No sweep cell of the epsilon-greedy rule has a matched opponent, and a softer rule also moves the level-1 model toward the level-0 opponents it faces, so a gain cannot be assigned to the bound alone.

Supplement B Run Settings and Run History

B.1 LLM Settings of the Components

Native reasoning, also called native thinking, is the reasoning that the model does before it answers. A manual chain of thought is step by step reasoning that the prompt asks for in the answer.

The LLM likelihood ran in two regimes. In regime 1, used in the method grids on seed 65685 (Sections 6.2.2 and 6.3), native thinking was on and the LLM rated the options directly. In regime 2, used in the mixed series on seeds 910001 to 914001 (Section 6.4) and in the cross-game experiment of The Resistance (Section 9.2), native thinking was off and the LLM reasoned in a manual chain of thought before it rated.

Against mixed opponents in Tofu Kingdom, the memoryless NeSy(IG) agent wins 74.0% of 100 games in the first regime (Table 6.7) and 65.3% of 150 games in the second (Table 6.9). The games differ as well, so the gap of 8.7 points cannot be assigned to the regime.

Table B.1. LLM settings of the components that call the model.

Component	Used in	Native thinking	Manual chain of thought
LLM Prince and LLM players (Tofu Kingdom)	LLM agent, NeSy(LLM) question choice, LLM opponents	high	none
LLM likelihood, regime 1 (Tofu Kingdom, method grids)	NeSy(IG) and NeSy(LLM) with the LLM likelihood	high	none
Portraits of the LLM type summary (Tofu Kingdom, mixed series)	one call per opponent after each completed game	none	none
LLM likelihood, regime 2 (Tofu Kingdom, mixed series)	memoryless and memory LLM likelihood; LLM behaviour type of Profile-mixture and Profile-joint	none	used (two calls per situation)
EMO (Tofu Kingdom)	opponent model and validator sub-calls	none	none
EMO decision call	questions and final guess	high	chain-of-thought module
GRAIL (Tofu Kingdom)	LLM prior of the factor graph	none	none
LLM player (The Resistance)	LLM player of the cross-game experiment	none	used
LLM likelihood, regime 2 (The Resistance)	NeSy(IG); LLM type of Profile-joint	none	used
old runs (Tofu Kingdom)	all LLM components	on (a thinking switch sent with each request; unlike the options in Supplement B.4, it took effect)	none

A call is repeated when its reply fails the schema check, and the seed is shifted by one for each attempt, so that a rejected reply is not generated again.

B.2 Parameters of Our Own Agents

Table B.2. Parameters of our own agents.

Parameter	Symbol	Tofu Kingdom	The Resistance	Defined in	How it was set
Response rule	none	default softmax; final experiments epsilon-greedy	softmax	Section 5.2 and Supplement A	design choice, compared and chosen in Supplement A
Error rate of the epsilon-greedy model	ε	default 0.1; final 0.2	not used	Eq A.1, Section 5.2 and Supplement A	chosen from the ε sweep (Table A.1)
Sharpness of the softmax model	β	default 8.0; not used in the final experiments	2.5, default and final	Section 5.2; softmax defined in Section 8.3.2	Tofu Kingdom: swept (Table A.2). The Resistance: fixed before the experiments
Weight of the Other faction in an answer value	λ	0.5 (not tuned)	not used	Eq 5.10b, Section 5.5	fixed value

Parameter	Symbol	Tofu Kingdom	The Resistance	Defined in	How it was set
Rating weights and confidence of the codebook	ω, ρ	$\omega = 4 : 2 : 1$; $\rho =$ 0.3, 0.6, 0.9	same	Eq 5.3, Section 5.2	fixed value
Hide weight	η	not used	default 0.5 for the level-1 model and above, 0 for beliefs with no model one level below; level-2 players 0.25; NeSy(IG) and Profile-joint 0	Eq 8.14, Section 8.3.2	final run settings
Information weight	γ	not used (plain information gain)	default 0.0; final 0.5 for NeSy(IG), Profile-joint and level-2 players	Eq 8.3, Section 8.2.1	final run settings
Adaptive information weight	none	not used	default false; final true, $\gamma_t = \gamma \times (1 - \Delta)$	Section 8.2.1	final run settings
Constants of the constants model	α, σ	not applicable	spy Fail 0.8, resistance Fail 0.0, spy approves spy team 0.8, spy steers 0.8	Eq 8.11 and 8.12, Section 8.3.1	fixed value
Constants of the random model	α, σ	not applicable	spy Fail 0.5, approve 0.5, steer 0.0, resistance Fail 0.0	Eq 8.11 and 8.12, Section 8.3.1	fixed value
Prune threshold for behaviour types	none	1e-6	1e-6	Sections 5.4 and 8.4	fixed value
Prior over behaviour types	none	uniform	uniform	Sections 5.4 and 8.4	default and final

Fixed value: set once and not varied in any experiment. Final run settings: the settings of the final Resistance runs.

The Resistance column shows the default and then the value of the final experiments, where they differ.

The level-2 behaviour type inside Profile-joint of The Resistance uses the same settings as the level-2 players.

B.3 Base Seeds and Notes

The last column points to the notes. Notes 5 and 6 apply to several rows.

Table B.3. Base seeds of the runs.

Used in	Game	Base seeds	Games per cell	Paired	Note
Supplement A and Sections 6.2 and 6.3 (old runs)	Tofu Kingdom	81 different base seeds	10 to 100	only a few cells	1
Sections 6.2 and 6.3	Tofu Kingdom, 7 players, center card	65685	100 (96 to 100 finished)	all cells	2

Used in	Game	Base seeds	Games per cell	Paired	Note
Section 6.4 and the cells (b) of Table 6.7	Tofu Kingdom, 7 players, center card	910001, 911001, 912001, 913001, 914001	30 per series, 150 in total	all arms	3
Section 9.2	The Resistance, 6 players	910001 to 915001, spaced 1000	30 per series, 6 series	the three arms	4

Notes to Table B.3.

1. Old runs. The configuration sweep of Supplement A.1 (5 players, no center card) has 64 base seeds, $991 + 10000(k-1)$ for k = 1 to 64, with 10 games each. 639 of the 640 games finished, and the 640 game seeds are all different. There are four full-scale checks of 100 games each (7 players, center card). Checks 1 and 3 use base seed 54628 and have identical role assignments game by game. Check 2 uses 4740099 and check 4 uses 45. The early cross-agent runs of the workshop paper (cells (a) of Table 6.7) use 124, except the LLM Prince against LLM opponents, which is check 3. The ablation of Table 6.3 has 12 base seeds, 10000 to 120000. A 5-player rerun of checks 1 and 3 used 946502 (75 of 100 games each, not in the old paper). This gives 64 + 3 + 1 + 1 + 12 = 81 base seeds. The game seeds 46 to 145 of check 4 and 125 to 224 of the cross-agent runs share 21 seeds. The random baseline of Supplement A.1 (base seed 76233, 100 games) is not among the 81.
2. Method grids. All cells play the same 100 games (game seeds 65686 to 65785). The main grid has 36 cells and its extension with the level-1 model has 18. The consistency grid (the level matrix of Section 6.3) has 9, the EMO cells 5 and the GRAIL cells of the method grids 3. Two more cells are NeSy(LLM) against level-1 opponents and against the mixed table. The nine-cell ablation of Section 6.2 completes the list of the cells used in Sections 6.2 and 6.3. The tuning sweeps of Supplement A.2 also use 65685.
3. Mixed series. Nine memory and baseline families share these 150 game seeds. The Sym(lv.1) arm was launched with 100 games per series by mistake, and we use games 0 to 29 of each series. The executed opponent order (seats 1 to 6) is level-0, LLM, level-0, LLM, level-2, level-2, with the opponents keeping their seats in all 45 older configurations. The arm of Section 6.4 with a level-1 opponent (the unmodelled arm) has it at seat 6. The configured order of the raw behaviour memory, Profile-mixture and LLM type summary arms was level-0, LLM, level-2, level-0, LLM, level-2, which does not match the runs.
4. Cross-game experiment of The Resistance. The spy seats are the same in all 180 pairs of games, and the series share no game. The two Spy cards go to two of the five other seats, and each of the ten pairs is equally likely. The first leader is uniform over the six seats. Both draws come from streams of the game seed. The tested agent is never a spy, so the absolute win rates and accuracies are not comparable with games in which all roles are assigned uniformly. The GRAIL arm ran on an earlier version of the code. The later changes to the code of the other players are refactors, input checks and log labels.
5. Interrupted batches. (a) The extension of the main grid: the 6 cells with level-0 opponents are complete, and the 7th cell stopped at 35 of 100 games. That cell and the 11 others were played again in full with seed 65685, and the merged table has 18 cells (1,800 games, 1,798 finished). (b) The consistency grid: one cell (level-1 opponents against the Sym(lv.2) Prince) stopped at 55 of 100 games and was played again two days later. (c) An earlier version of the Profile-mixture family stopped at the first game of its fourth series. It was replaced and is not used. Not interruptions: the Profile-joint family of Section 6.4 was started in three launches with unchanged seeds, and no run loaded a stored memory.
6. Random streams. The seed tree of Section 6.1 holds for the runs from June 2026 on. The LLM prior of the Tofu Kingdom GRAIL analogue is not seeded.

B.4 Run History

The old runs sent a context of 8192 tokens, an output limit of 4096 tokens and a repeat penalty of 1.1 or 1.2 with each request. The /v1 endpoint of Ollama ignores these options. So the old runs used the unmodified model at the default context of the server, with no output cap from the client and with the Ollama default repeat penalty of 1.1.

The later runs use the model variant qwen3.5-9b-ctx12288, which changes only the context length, so all runs use the default repeat penalty of 1.1. In 400 games in which the LLM made only the final guess, the prompt of that call had 1,651 to 4,098 tokens, with a median of 1,964. Generated tokens also count against the context window, and we did not measure them.

Table B.4. Software and hardware of the two experiment machines.

Item	Machine A	Machine B
GPU	NVIDIA L40S, 24 GB vGPU partition of a 48 GB card	NVIDIA L40S, 12 GB vGPU partition
Driver and CUDA	580.126.09, 13.0	580.126.09, 13.0
CPU and memory	16 vCPU, about 118 GB	8 vCPU, 59 GiB
Operating system	Ubuntu 22.04.5 LTS, kernel 5.15.0-185	Ubuntu 22.04.5 LTS, kernel 5.15.0-194
Ollama and Python	0.18.2, 3.12.13	0.18.2, 3.12.13
openai, instructor, pydantic	2.29.0, 1.14.5, 2.12.5	same
numpy, torch	2.4.6, 2.11.0+cu130	same
Tofu Kingdom runs	all except the unmodelled arm	the unmodelled arm
Resistance runs	GRAIL: series 0 to 5. NeSy(IG): series 0 and 1. Profile-joint: series 0 to 3	NeSy(IG): series 2 to 5. Profile-joint: series 4 and 5

Some experiments shared one inference server, so their wall clock time includes waiting in a queue. In The Resistance, the NeSy(IG) arm moved to the smaller machine B after series 1, and its time per game rose from about 700 to about 1100 seconds.

The crash rates in the cross-game experiment of The Resistance are 3.3% for GRAIL, 11.7% for the NeSy(IG) arm and 9.4% for the Profile-joint arm. Every crash is an LLM call that failed all 3 attempts. The six crashes of the GRAIL arm are closing queries of the LLM player, made after the game was decided, because the GRAIL agent makes no LLM call. The winner was already recorded in 10 of the 44 crashed games (GRAIL 6, NeSy(IG) 1, Profile-joint 3), and these games are excluded as well.

In Tofu Kingdom the same failure removed up to 4 of 100 games in a cell (EMO and the LLM Prince against the mixed table) and 5 and 3 of 150 games of the raw behaviour memory and LLM type summary arms.

The Profile-joint arm of Table 6.9 ran on machine A in July 2026 and the unmodelled arm on machine B in October 2026. Both machines report Ollama 0.18.2 and the same package versions (Table B.4), and both arms use the model variant qwen3.5-9b-ctx12288.

No agent in the reported runs revises a team, so the limit on proposals before a vote (Section 7.1) never binds in them. At the limit the game goes on, unlike on the GRAIL server, because ending games in which an LLM stalls would bias the sample.

B.5 Notes on the Statistics

The bootstrap intervals for identity accuracy in The Resistance (Section 9.1) use 20,000 resamples with a fixed seed.

B.6 Adaptation of the Baselines

GRAIL in The Resistance. The code is the Graph Only version in the repository of the authors (Bayesian-Avalon, MIT licence, commit 1f1faaa) [2]. We kept its factor graph, its belief propagation, its

trained neural model file and its policy. The model file is the one that the released code loads, copied without change. Our environment (Section 7.1) calls the agent directly instead of through the authors' game server.

One change to the released code affects which player is drawn in a given game, but not the distribution of the draws. The policy fills a team of two with a random second member. The original draws it from a global random generator that none of our seeds reaches. We draw it from the seeded stream of Section 6.1.

Two behaviours of the original were not copied, because our environment has no discussion. The original updates its belief after every chat message. Our agent updates it after each mission. In the released games, 129 of 324 proposals were revisions after the discussion. Our leader calls the vote after its first proposal.

The leader puts itself on the team and adds the players who look most like resistance players. For a team of two, the other member is drawn at random. A GRAIL player votes yes on any team of two. Otherwise it votes yes unless some member of the team looks more like a spy than a resistance player. On the fifth proposal of a round, after four rejections, it votes yes.

GRAIL in Tofu Kingdom. Section 5.6 describes our analogue. The graph is rebuilt from the answers on every inference. The LLM prior is used only after at least one answer exists, and it is not seeded (note 6 of Supplement B.3). The network behind the learned factors is trained on games against LLM opponents. The evaluation games also have level-0 and level-2 opponents, so on those the learned factors meet opponents that they were not trained on.

EMO. Table B.1 gives the LLM settings of its calls. The main EMO runs have reasoning history on, as the LLM agents do. A short check against LLM opponents used the first 20 games of the series of the main runs (seed 65685) and did not choose any setting. EMO wins 11 of 20 games with reasoning history on and 6 of 19 with it off (one game crashed). The two cannot be separated (exact McNemar p = 0.29 on the 19 paired games).

Supplement C The Simplified Bayes Update of Tofu Kingdom

Equation 5.1 leaves out the factor $P(q \mid w, c)$ of the full Bayes update. The Prince's question-selection procedure is determined by his belief $P(w \mid c)$, which is itself determined by c alone. The Prince has no access to the true world w^*, so he cannot condition on it. Therefore, the question choice is conditionally independent of w^* given c:

$$P(q \mid w, c) = P(q \mid c) = \kappa \quad \text{for all accessible } w.$$

Substituting this into the full Bayes update gives

$$P(w \mid r, q, c) \propto P(r \mid w, q, c)\, P(q \mid w, c)\, P(w \mid c) = P(r \mid w, q, c)\, \kappa\, P(w \mid c) \propto P(r \mid w, q, c)\, P(w \mid c),$$

where κ cancels in the normalization. This is precisely Equation 5.1. The constant κ never needs to be computed.

References

[1] Haochong Luo, Guang Hu, and Chenyuan Zhang. Active information gathering in social deduction game. In *Proceedings of the ICAPS 2026 Workshop on Roles of Planning in Games (RPG)*, June 2026.

[2] Shahab Rahimirad, Guven Gergerli, Lucia Romero, Angela Qian, Matthew Lyle Olson, Simon Stepputtis, and Joseph Campbell. Bayesian social deduction with graph-informed language models. *ArXiv*, abs/2506.17788, 2025.